

Schur Rings over Semigroups

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For a group G , we will identify finite subsets $C \subseteq G$ with elements of the group algebra $\mathbb{Q}[G]$, namely the **simple quantity** $\sum_{g \in C} g \in \mathbb{Q}[G]$.

For $C \subseteq G$, let $C^{(-1)} = \{g^{-1} \mid g \in C\}$.

Likewise, if $\mathfrak{S}$ be a partition of G of **finite support**, meaning all classes in $\mathfrak{S}$, called **primitive sets**, are finite. Then we may identify $\mathfrak{S}$ with a subspace of $\mathbb{Q}[G]$, namely

$$\mathbb{Q}[\mathfrak{S}] := \text{Span}_{\mathbb{Q}}(\mathfrak{S}) = \text{Span}_{\mathbb{Q}}\{C \mid C \in \mathfrak{S}\},$$

that is, the subspace spanned by the primitive sets in $\mathfrak{S}$.

For a group G , we will identify finite subsets $C \subseteq G$ with elements of the group algebra $\mathbb{C}[G]$, namely the **simple quantity** $\sum_{g \in C} g \in \mathbb{C}[G]$.

For $C \subseteq G$, let $C^{(-1)} = \{g^{-1} \mid g \in C\}$.

Likewise, if $\mathfrak{S}$ be a partition of G of **finite support**, meaning all classes in $\mathfrak{S}$, called **primitive sets**, are finite. Then we may identify $\mathfrak{S}$ with a subspace of $\mathbb{C}[G]$, namely

$$\mathbb{C}[\mathfrak{S}] := \text{Span}_{\mathbb{C}}(\mathfrak{S}) = \text{Span}_{\mathbb{C}}\{C \mid C \in \mathfrak{S}\},$$

that is, the subspace spanned by the primitive sets in $\mathfrak{S}$.

Let $\mathfrak{S}$ be a partition of G of finite support. We say $\mathfrak{S}$ is a **Schur Ring** (or *S-ring*) if

- $CD = \sum_{E \in \mathfrak{S}} \lambda_E E \in \mathbb{C}[\mathfrak{S}]$, for all $C, D \in \mathfrak{S}$,
- $\{1\} \in \mathfrak{S}$,
- $C^{(-1)} \in \mathfrak{S}$, for all $C \in \mathfrak{S}$.

Association Schemes

In 1985, Klin showed that Schur rings can be used to form association schemes:

The group algebra $\mathbb{C}[G]$ coincides to the association scheme $\mathcal{A} = \{R_g \mid g \in G\}$ on the set G by the relations

$$R_g = \{(h, k) \mid h = gk\}.$$

More generally, a Schur ring $\mathbb{C}[\mathfrak{G}] \subseteq \mathbb{C}[G]$ coincides to the association scheme $\mathcal{A} = \left\{ \bigcup_{g \in C} R_g \mid C \in \mathfrak{G} \right\}$.

A **semigroup** $(G, *)$ is a magma obeying the *associative property*, that is, for each $a, b, c, \in G$:

$$a(bc) = (ab)c.$$

A semigroup with an identity is called a **monoid**.

Schur Rings over a Semigroup

Definition

Let $(G, *)$ be a semigroup, and let $\mathfrak{S}$ be a partition of G of finite support. Let

$$\mathbb{C}[G] = \left\{ \sum_{\alpha \in \mathcal{A}} c_{\alpha} \alpha \mid c_{\alpha} \in \mathbb{C} \right\} = \text{Span}\{\alpha \mid \alpha \in \mathcal{A}\}$$

be the semigroup algebra with complex coefficients.

Let $\mathbb{C}[\mathfrak{S}] = \text{Span}\{C \mid C \in \mathfrak{S}\}$. Then we say that $\mathfrak{S}$ is a **Schur ring** over G if

- $CD = \sum_{E \in \mathfrak{S}} \lambda_E E \in \mathbb{C}[\mathfrak{S}]$, for all $C, D \in \mathfrak{S}$,
- $\{1\} \in \mathfrak{S}$,
- $C^{(-1)} \in \mathfrak{S}$, for all $C \in \mathfrak{S}$.

Schur Rings over a Monoid

Definition

Let $(G, *)$ be a monoid, and let $\mathfrak{S}$ be a partition of G of finite support. Let

$$\mathbb{C}[G] = \left\{ \sum_{\alpha \in \mathcal{A}} c_{\alpha} \alpha \mid c_{\alpha} \in \mathbb{C} \right\} = \text{Span}\{\alpha \mid \alpha \in \mathcal{A}\}$$

be the monoid algebra with complex coefficients.

Let $\mathbb{C}[\mathfrak{S}] = \text{Span}\{C \mid C \in \mathfrak{S}\}$. Then we say that $\mathfrak{S}$ is a **Schur ring** over G if

- $CD = \sum_{E \in \mathfrak{S}} \lambda_E E \in \mathbb{C}[\mathfrak{S}]$, for all $C, D \in \mathfrak{S}$,
- $\{1\} \in \mathfrak{S}$,
- $C^{(-1)} \in \mathfrak{S}$, for all $C \in \mathfrak{S}$.

Schur Rings over Left Null Semigroups

A **left null semigroup** of order n , denoted LO_n , is the semigroup such that for all $x, y \in LO_n$ we have

$$xy = x.$$

Theorem

Every partition of LO_n is a Schur ring.

Schur Rings over Chains

A **chain** of order n , denoted CH_n , is the semigroup such that $CH_n = \{1, 2, \dots, n\}$ and the operation is max.

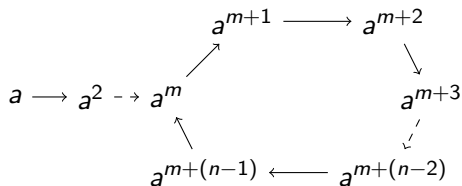
Theorem

The discrete partition is the only Schur rings over CH_n .

Schur Rings over Cyclic Semigroups

A **monogenic semigroup** is a semigroup given by the presentation

$$\mathcal{Z}_{m,n} = \langle a \mid a^m = a^{m+n} \rangle.$$



Note that when $m = 1$, $\mathcal{Z}_{m,n} \cong \mathcal{Z}_n$, the cyclic group of order n .

Theorem

If $m > 1$, then the only Schur rings over $\mathcal{Z}_{m,n}$ is discrete.

Indecomposable Elements

Let G be a semigroup and let $a \in G$. We say that a is **decomposable** if there exists $b, c \in G$ such that

$$a = bc.$$

Otherwise, we say that a is **indecomposable**.¹

Theorem

Let $\mathfrak{S}$ be a Schur ring over a semigroup G . Let $a \in G$ be indecomposable. Then the primitive set containing a only contains indecomposable elements.

¹We consider idempotent elements x likewise indecomposable if their only factorization is $x = x^2$.

Indecomposable Elements

Let $\mathcal{I}_0$ be the set of indecomposable elements of $G_0 = G$. Let $G_1 = G_0 \setminus \mathcal{I}_0$. This is an ideal of G_0 and hence a semigroup. Let $\mathcal{I}_1$ be the set of indecomposable elements of G_1 .

Recursively, define $\mathcal{I}_n$ be the set of indecomposable elements in G_n , for which $G_n = G \setminus \bigcup_{k=0}^{n-1} \mathcal{I}_k$ is the ideal of decomposable elements in G_{n-1} . We call the elements of $\mathcal{I}_n$ the **n -decomposable elements** of G .

This decomposition offers a **layering** of the semigroup:

$$G = \mathcal{I}_0 \cup \mathcal{I}_1 \cup \mathcal{I}_2 \cup \dots \cup G_\infty$$

Theorem

Every Schur ring over a semigroup is a refinement of the above layering.

Schur Rings over Semilattices

A **semilattice** is a magma which is commutative, associative, and all elements are idempotent:

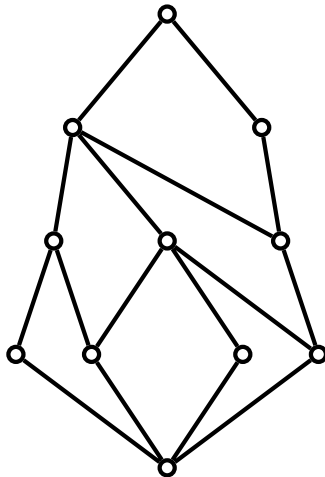
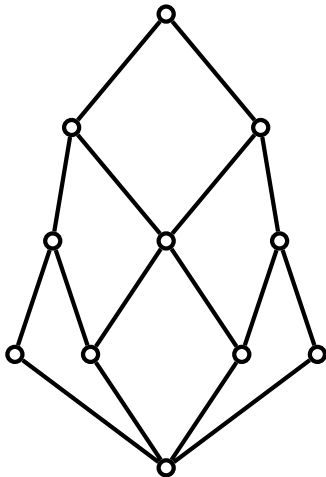
$$x^2 = x.$$

Let L be a semilattice and let $a \in L$. Then the **root** of a is $\lfloor a \rfloor := \{x \in L \mid x \preceq a\}$.

Theorem

The partition over semilattice induced by the equivalence relation $\lfloor a \rfloor \cong \lfloor b \rfloor$ is a Schur ring.

Schur Rings over Semilattices



Why Do Schur Rings over Semigroups Matter?

Let G be a finite group and let L be a lattice of normal subgroups of G such that $1, G \in L$. Then

$$\mathfrak{S} = \text{Span}_{\mathbb{C}}\{H \mid H \in L\}$$

is a Schur ring over G , called a **lattice Schur ring**.

Now,

$$\mathbb{C}[\mathfrak{S}] \cong \mathbb{C}[L].$$

As this isomorphism is combinatorial, any Schur ring over L is also a Schur subring of $\mathbb{C}[\mathfrak{S}]$, and hence a Schur ring over G .

Hence, understanding Schur rings over semilattices will be necessary to classify Schur rings over elementary abelian groups.

Thank you for listening!